Two-channel signatures of the E_7 spectrum in a blockaded Rydberg chain

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The thermal deformation of the tricritical Ising conformal field theory has an integrable E_7 spectrum with seven particles. We investigate its light-particle signatures in a strictly blockaded Rydberg-chain model with an attractive $1/r^6$ interaction tail. Uniform and staggered density probes select the even and odd sectors of the translation-invariant low-energy theory. Exact diagonalization up to $L = 36$ identifies a direction-dependent region where three finite-chain gap ratios agree with those of A_1–A_4 to within 3%. Open-chain spectra retain four candidate resonances, but a smooth drive envelope suppresses channel mixing unevenly: at $L = 27$ the leakage is below 0.03% for A_1, A_2, but about 7.5% and 1.9% for A_3, A_4. Periodic spectra are consistent with the expected two-particle onsets; they do not determine threshold exponents. The three heavy particles remain unresolved. Shallow-bound-state corrections and experimental resolution remain to be controlled.

Introduction.—Relevant perturbations of a two-dimensional conformal field theory (CFT) can produce integrable massive theories with spectra fixed by exceptional Lie algebras. The magnetic perturbation of critical Ising theory realizes E_8 [1], with experimental signatures in $CoNb_2O_6$ [2] and Rydberg arrays [3]. The leading thermal perturbation of the tricritical Ising (TCI) CFT, with central charge $c = 7/10$, instead realizes E_7 [4–6]. Writing $r_a = m_a/m_1$, the seven ratios are

$$(r_1, \ldots, r_7) = (1,\ 1.2856,\ 1.8794,\ 1.9696,\ 2.5321,\ 2.8794,\ 3.7017). \quad (1)$$

with $\mathbb{Z}_2$ parities $(-,+,-,+,+,-,+)$ in the disordered phase [7]. An experimental identification of this spectrum remains an open problem [8].

Rydberg chains support period-two order and Ising criticality, and an effective attraction can produce a tricritical point [9, 10]. Sun *et al.* [11] reached the TCI regime by preparing the upper edge of the blockaded sector and measured finite-size CFT spectra. In the projected model, reversing the Hamiltonian and the sign of the transverse drive maps this preparation to an attractive ground-state problem. This motivates spectroscopy of the neighboring gapped regime.

The field-theory inputs are established: the Rydberg density–CFT operator correspondence [10], the E_7 form factors and selection rules [6, 12], and their threshold structure. Király and Lencsés [13] have also obtained lattice-quench spectra compatible with the even particles A_2, A_4, A_5 in the Blume–Capel chain, including form-factor amplitude comparisons. That study uses A_2 to calibrate the mass scale; it does not independently resolve A_1. Here we compare both light-particle sectors of a Rydberg model and quantify finite-chain parameter dependence and boundary mixing.

Model and field content.—We study the projected Hamiltonian

$$H = \sum_j \left[\frac{\Omega}{2} P_{j-1} X_j P_{j+1} - \Delta n_j\right] + \sum_{i<j,\, d_{ij}\geq 2} V_2 (2/d_{ij})^6 n_i n_j. \quad (2)$$

with the constraint $n_j n_{j+1} = 0$, $P_j = 1 - n_j$, and $V_2 < 0$. The distance d_{ij} is $|i-j|$ for open chains and the shortest distance on a periodic chain; each pair is counted once. We set $\hbar = \Omega = a = 1$, where a is the lattice spacing. All allowed interaction pairs are retained within each finite chain.

Equation (2) is the strict-blockade approximation to the sign-reversed experimental problem, not the full experimental Hamiltonian. It omits the finite-blockade correction H_2 of Ref. [11], which contains diagonal shifts and correlated hopping of order Ω^2/V_1. Our open chains have no additional boundary detuning profile. We compute ground-state response, without simulating the preparation sweep, leakage outside the blockade sector, or readout.

In the long-wavelength theory, one-site translation exchanges the two density-wave patterns and implements the internal $\mathbb{Z}_2$ together with spatial translation [10]. The density has the expansion

$$n_j \simeq \bar{n} + c_\varepsilon \varepsilon + c_{\varepsilon'} \varepsilon' + (-1)^j (c_\sigma \sigma + c_{\sigma'} \sigma') + \cdots . \quad (3)$$

Thus zero-momentum even particles occur near lattice momentum 0, and odd particles near π. This identification uses bulk translation symmetry; open-chain reflection parity is a different quantum number.

Both Δ and V_2 generate the even relevant perturbations $g_\varepsilon \varepsilon + g_{\varepsilon'} \varepsilon'$, with RG eigenvalues 9/5 and 4/5. For a straight path of distance t from TCI with a nonzero linear component in g_ε,

$$\eta = \frac{g_{\varepsilon'}}{|g_\varepsilon|^{4/9}} \propto t^{5/9}, \qquad m_1 \propto t^{5/9}. \quad (4)$$

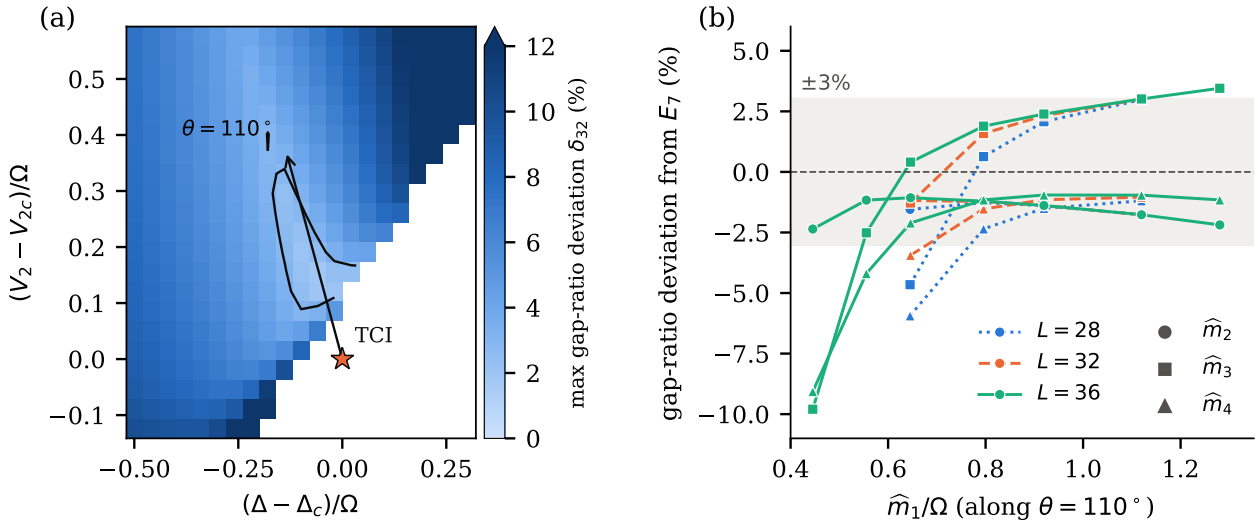


FIG. 1. Finite-chain spectral matching region. (a) $\delta_L = 100\,\max_{a=2,3,4}|(\widehat{m}_a/\widehat{m}_1)/r_a - 1|$ at $L = 32$; the contour marks 3%. Grid spacings are 0.04Ω in Δ and 0.035Ω in V_2. Blank cells fail the numerical filters $L\widehat{m}_1/\Omega \geq 18$ and $\widehat{m}_2 \geq \widehat{m}_1/2$; These filters do not establish phases or convergence. The star estimates TCI; the arrow marks $\theta = 110°$. (b) Individual gap-ratio deviations along that direction. Symbols identify the candidate levels and line styles identify sizes. The band is ±3%; lines guide the eye. Neither panel is an experimental error budget.

On the disordered side such paths approach the thermal E_7 theory as $t \to 0$ if the system size is increased simultaneously. The growing correlation length limits this approach at fixed L. We have not calibrated the map from lattice parameters to $(g_\varepsilon, g_{\varepsilon'})$; a region of small mass-ratio errors need not identify the pure thermal trajectory or separate competing corrections.

Probes and numerical method.—For a periodic ground state, the connected density response is

$$S_k(\omega) = \frac{1}{L}\sum_{n\neq 0}\left|\left\langle n\left|\sum_j e^{ikj}n_j\right|0\right\rangle\right|^2 \delta[\omega-(\mathcal{E}_n-\mathcal{E}_0)]. \quad (5)$$

Uniform and staggered probes select $k = 0$ and π exactly. In the E_7 limit, Eq. (3) couples them principally to ε and σ, respectively. The proposed $k = \pi$ coupling to σ already appears in Ref. [11]. For a pure continuum operator $\mathcal{O}$, one-particle weights obey $w_a \propto |F_a^{\mathcal{O}}|^2/m_a$. Known form factors give $w_3/w_1 = 0.067$, $w_6/w_1 = 0.0030$ for σ, and $w_4/w_2 = 0.143$, $w_5/w_2 = 0.038$, $w_7/w_2 = 0.0012$ for $\Theta \propto \varepsilon$ [6, 12]. Absolute interchannel amplitudes are not universal.

We use sparse exact diagonalization in the constrained space for periodic $L \leq 36$ and open $L = 19, 23, 27$. Spectra use reorthogonalized Lanczos iterations (up to 400 for the periodic spectra and 250 for the open spectra shown). Small-system checks compare momentum-sector energies and weights with full-space diagonalization to numerical precision. They do not establish convergence of the large-system continuum.

We denote the first two gaps in the $k = \pi$ sector by $\widehat{m}_1(L), \widehat{m}_3(L)$ and the first two excited gaps at $k = 0$ by $\widehat{m}_2(L), \widehat{m}_4(L)$. These are candidate light-particle levels, not independently extrapolated masses. Throughout the parameter scan, the labels follow energy ordering within each sector. For open chains, ω_a^{o} denotes a candidate resonance energy selected by its dominant probe weight.

Simultaneous crossings of L times the lowest odd and lowest excited even gaps locate a finite-size TCI estimate. The $L = 28/32$ crossing gives $(\Delta_c, V_{2c}) = (-1.5011, -1.6260)$, within 10^{-3} of the 24/28 estimate and close to the experimental calibration [11]. Fixing $v \simeq 1.90$ using $x_\sigma = 3/40$ gives other rescaled gaps near 0.875 at $k = \pi$, and 0.198, 1.18 at $k = 0$, consistent with $x_{\sigma'} = 7/8$, $x_\varepsilon = 1/5$, $x_{\varepsilon'} = 6/5$. The value of x_σ is an input to this comparison.

Matching region and finite size.—We scan $(\Delta, V_2) = (\Delta_c, V_{2c}) + t(\cos\theta, \sin\theta)$. Figure 1 identifies a finite region of small errors among the three gap ratios. Along $\theta = 110°$, the sampled $L = 36$ points at $t = 0.1, 0.15, 0.2$ have maximal errors below 3%; at $t = 0.3$ the maximum is 3.01%. Nearby directions 90° and 130° also contain matching points. In contrast, the sampled $\theta = 180°$ points never reach 5% accuracy; deviations near 210° can reach 10–25%. This direction dependence concerns the accessible finite-chain window and does not contradict the generic asymptotic limit in Eq. (4).

At $\theta = 110°, t = 0.2$, corresponding to $(\Delta, V_2) = (-1.5695, -1.4381)$, we find $\widehat{m}_1(36) = 0.9196\Omega$ and

$$\frac{(\widehat{m}_2, \widehat{m}_3, \widehat{m}_4)}{\widehat{m}_1} = (1.2678,\ 1.9243,\ 1.9508). \quad (6)$$

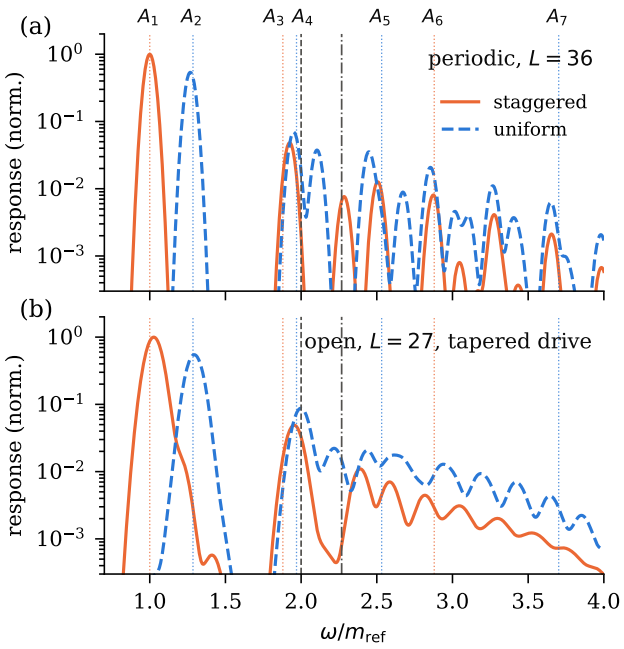


FIG. 2. Response at Eq. (6): (a) periodic $L = 36$, without size averaging; (b) open $L = 27$ with a $\sin^2$ drive envelope. Both use $m_{\rm ref} = \widehat{m}_1(36) = 0.9196\Omega$ and are normalized to their own staggered peak height. Solid orange: staggered; dashed blue: uniform. Only (a) has exact momentum selection. Gaussian standard deviations $0.03m_{\rm ref}$ and $0.05m_{\rm ref}$ illustrate smoothing, not instrumental linewidths. Dotted lines mark ideal E_7 ratios. Gray dashed/dash-dotted lines mark $2m_{\rm ref}$ and $m_{\rm ref} + \widehat{m}_2(36)$: periodic-gap sums, not measured open-chain continuum edges.

The relative deviations are $(-1.39, +2.39, -0.95)\%$. At $t = 0.1$, however, $\widehat{m}_3/\widehat{m}_1$ changes from 1.792 to 1.887 between $L = 28$ and 36, illustrating the limitation closer to TCI. At $t = 0.3$ the three ratios change by less than 0.24% over those sizes, but this comparison is not an infinite-volume extrapolation.

The shallow A_4 bound state requires particular care. In the integrable theory, A_4 occurs in the A_1A_1 scattering channel, with

$$\mu_{11}^4 = \sqrt{m_1^2 - m_4^2/4} = m_1 \sin(\pi/18) \simeq 0.174 m_1. \quad (7)$$

Its finite-volume energy can receive a leading bound-state correction proportional to $e^{-\mu_{11}^4 L/v}$, not simply $e^{-m_1 L/v}$ [12, 14]. Using the working gap and critical velocity as estimates gives $v/\mu_{11}^4 \simeq 12$ sites, compared with $v/m_1 \simeq 2$ sites. This is an E_7-limit estimate, not a fitted correction to our nonintegrable chain. In the data, $\widehat{m}_4/\widehat{m}_1$ changes from 1.9400 to 1.9508 between $L = 28$ and 36: only 0.56% of its value, but 22% of the latter binding-energy estimate $2\widehat{m}_1 - \widehat{m}_4$. A mass-ratio tolerance alone therefore does not control near-threshold binding.

At the working point, the periodic weight ratios are $w_4/w_2 = 0.129$ and $w_3/w_1 = 0.0483$. Mixing with σ' contributes an effective relative amplitude proportional to $(c_{\sigma'}/c_\sigma)(m_1 a/v)^{4/5}$ at fixed CFT normalization, since $x_{\sigma'} - x_\sigma = 4/5$. The weights also contain interference, finite-size, and nonintegrable corrections. We do not isolate these sources or treat the pure-field ratios as precise lattice predictions.

Spectra and boundaries.—The periodic response in Fig. 2(a) has two prominent subthreshold peaks in each sector, consistent with the four candidate levels in Eq. (6). Momentum selection is exact. The next visible even-sector level at $L = 36$ is at $2.105m_{\rm ref}$, so its separation from the A_4 candidate is $0.154m_{\rm ref}$, appreciably larger than the gap to the kinematic reference $2m_{\rm ref}$. Discrete scattering levels can thus make a small-system spectrum appear more separated than its infinite-volume counterpart.

For open chains we use $O_{f,q} = \sum_j f_j e^{iqj} n_j$, where $q = 0, \pi$ labels the drive pattern rather than an exact momentum sector. Spectral weights are normalized by $\sum_j f_j^2$. We compare $f_j = 1$ with $f_j = \sin^2[\pi(j+1)/(L+1)]$, $j = 0, \ldots, L-1$. At $L = 27$, the latter gives dominant candidate resonance ratios

$$(\omega_2^{\rm o}, \omega_3^{\rm o}, \omega_4^{\rm o})/\omega_1^{\rm o} = (1.2612,\ 1.9030,\ 1.9488), \quad (8)$$

with $\omega_1^{\rm o} = 0.9438\Omega$. The ratios lie within 2% of E_7, but their denominator differs from $m_{\rm ref}$. In the common units of Fig. 2, $\omega_4^{\rm o} \simeq 2.0001 m_{\rm ref}$. Neither normalization alone establishes the binding energy of an open-chain resonance.

Table I reports all four channel leakages, defined as the integrated weight at the same candidate eigenenergy in the unintended probe divided by its weight in the intended probe. At $L = 27$, tapering reduces the A_1 leakage from 0.149% to 0.0267%, but the A_3 leakage only from 9.22% to 7.46%, and the A_4 leakage increases from 1.41% to 1.94%. Thus tapering does not uniformly separate the sectors. At $L = 19$, two nearby uniform resonances share substantial weight, making the assignment less clean than at $L = 27$. A further level at $1.160\,\omega_1^{\rm o}$ has staggered weight 1.7% of the A_1 candidate, down from 13% for a flat drive. Residual satellites and higher-particle leakage must be included in any experimental peak assignment.

Thresholds and unresolved heavy particles.—In the integrable, translation-invariant theory, the lowest even and odd two-particle thresholds are $2m_1$ (A_1A_1) and m_1+m_2 (A_1A_2). The exact E_7 scattering amplitudes and form factors imply responses proportional to $(\omega - 2m_1)^{1/2}$ and $(\omega - m_1 - m_2)^{-1/2}$ in the corresponding channels [12]. Here $S_{11}(0) = -1$ forces the identical-particle form factor to vanish, whereas the relevant A_1A_2 form factor is nonzero at threshold. These statements concern matrix-element-weighted response, not bare two-particle phase space, and are predictions of the integrable limit.

TABLE I. Leakage ratios in percent for tapered open-chain probes at the working point. The intended probe is staggered for A_1, A_3 and uniform for A_2, A_4. Labels denote finite-chain candidates.

Candidate	$L = 27$	$L = 19$
A_1	0.0267	0.228
A_2	0.0192	4.53
A_3	7.46	39.4
A_4	1.94	8.98

At our working point, the next odd-sector spectral level above the two light peaks moves from $2.340\widehat{m}_1$ at $L = 18$ to $2.284\widehat{m}_1$ at $L = 36$, toward the gap sum $1 + \widehat{m}_2/\widehat{m}_1 = 2.268$. The corresponding even-sector sequence moves from $2.296\widehat{m}_1$ at $L = 24$ to $2.105\widehat{m}_1$ at $L = 36$. This behavior is consistent with different two-particle onsets, but does not determine continuum line shapes or threshold exponents. In open chains, residual weight above the A_3 candidate prevents interpreting the corresponding interval as a strictly empty spectral window.

We do not resolve A_5, A_6, A_7. In the E_7 theory their masses lie above allowed thresholds in their own sectors: A_5, A_7 above $2m_1$, and A_6 above $m_1 + m_2$. Nonintegrable perturbations can open decay channels such as $A_5 \to A_1A_1$ and $A_6 \to A_1A_2$ [7]. Across $L = 18$–36 at $\theta = 110°$, $t = 0.2, 0.3$, the higher spectra do not provide a stable heavy-particle assignment by energy and weight. Finite closed-chain spectra do not measure decay widths; this outcome neither establishes nor excludes heavy resonances in larger systems or experiments.

Experimental requirements.—The demonstrated TCI preparation and spatial detuning control provide relevant starting capabilities [11]. A candidate protocol would prepare a state near Eq. (6), apply weak uniform and staggered modulations, and compare the two responses. However, the sign-changing staggered drive needed for a pure σ probe is not established by those capabilities alone. Reference [11] discusses the one-color light-shift constraint and proposes combining 515 and 813 nm tweezers as a future implementation. Unwanted static components and the tapered spatial profile would also need to be controlled.

Spectral resolution is especially consequential for A_4. The finite-chain estimate $2\widehat{m}_1 - \widehat{m}_4 \simeq 0.049m_{\rm ref}$ is about $2\pi \times 0.25$ MHz for $\Omega = 2\pi \times 5.5$ MHz; in ideal E_7 the relative binding is smaller, $0.0304m_1$. The illustrative open-chain broadening in Fig. 2 has FWHM $= 2\sqrt{2\ln 2}\,\sigma = 0.118m_{\rm ref}$, about $2\pi \times 0.60$ MHz. A visible local maximum under this smoothing does not establish separability from a continuum and noise. The E_7 threshold suppression may help in its scaling regime, but is not verified at this lattice working point. Likewise, open-chain mixing means that separating A_3 and A_4 cannot rely solely on ideal channel labels.

The quoted 1.7% uncertainty in V_2 [11] corresponds to roughly 0.028Ω near TCI, comparable to our 0.035Ω grid spacing. Figure 1 therefore does not certify robustness to that uncertainty. A joint error budget for Δ, V_2, Ω, boundary control, finite blockade, preparation, and detection remains outside this calculation.

Conclusion.—A strictly blockaded Rydberg-chain model exhibits four light spectral candidates with the expected two-channel organization and few-percent agreement with E_7 gap ratios in a finite parameter region. Open chains preserve recognizable resonances but have particle-dependent channel mixing. Periodic higher levels are consistent with the expected two-particle onsets; the three heavy particles remain unresolved. Establishing an experimental E_7 identification requires controlling the shallow A_4 bound state and validating the full driven, noisy response. The present results specify a model regime and its limitations rather than a completed observability demonstration.

[Acknowledgments.]

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